ITO / PELTIER CONTROLLER

24 V | XIAO nRF52840 | 4 TEC-kanalen | ITO-heater | 4 A prototype

VERMOGEN EN BEVEILIGING

02 R7B / ingang / eFuse

File: sheet_02.kicad_sch

03 Relais / noodstop

File: sheet_03.kicad_sch

04 Heater / stroommeting

File: sheet_04.kicad_sch

BESTURING EN STROOMINSTELLING

01 MCU / 5V / USB

File: sheet_01.kicad_sch

05 DAC / opampbuffers

File: sheet_05.kicad_sch

06 TEC-vrijgave

File: sheet_06.kicad_sch

VIER GELIJKE TEC-STROOMREGELAARS

07 TEC 1

File: sheet_07.kicad_sch

08 TEC 2

File: sheet_08.kicad_sch

09 TEC 3

File: sheet_09.kicad_sch

10 TEC 4

File: sheet_10.kicad_sch

TEMPERATUUR, VENTILATOR EN OMGEVING

11 Glas-NTCs / ADC

File: sheet_11.kicad_sch

12 Heatsink / fan / I2C

File: sheet_12.kicad_sch

13 Remote SHT31-poorten

File: sheet_13.kicad_sch

Bladen 02–14: echte bedrading binnen ieder blok. Gelijke voedingssymbolen / bladpoorten zijn elektrisch verbonden.

ERC-bronverklaringen voor aanwezige voedingsrails:

PWR_FLAG GND

PWR_FLAG +24_IN

PWR_FLAG +24_PROT

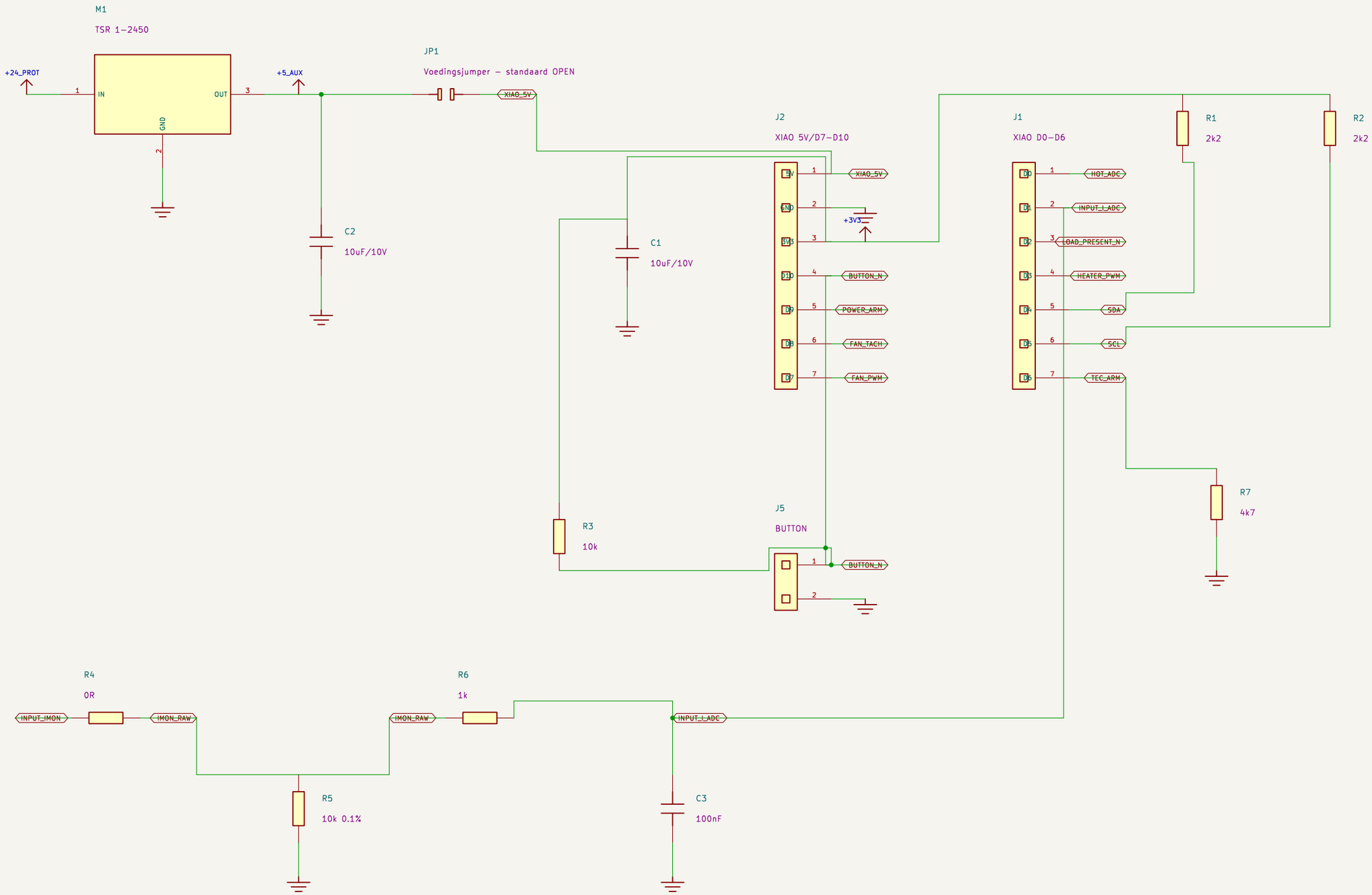
PWR_FLAG +24_LOAD

PWR_FLAG +5_AUX

PWR_FLAG +3V3

01 MCU en voeding van de logica

ITO controller L | 24 V | prototype 4 A | bedrading volgens bestaande PCB



02 Ingangsbeveiliging

ITO controller L | 24 V | prototype 4 A | bedrading volgens bestaande PCB

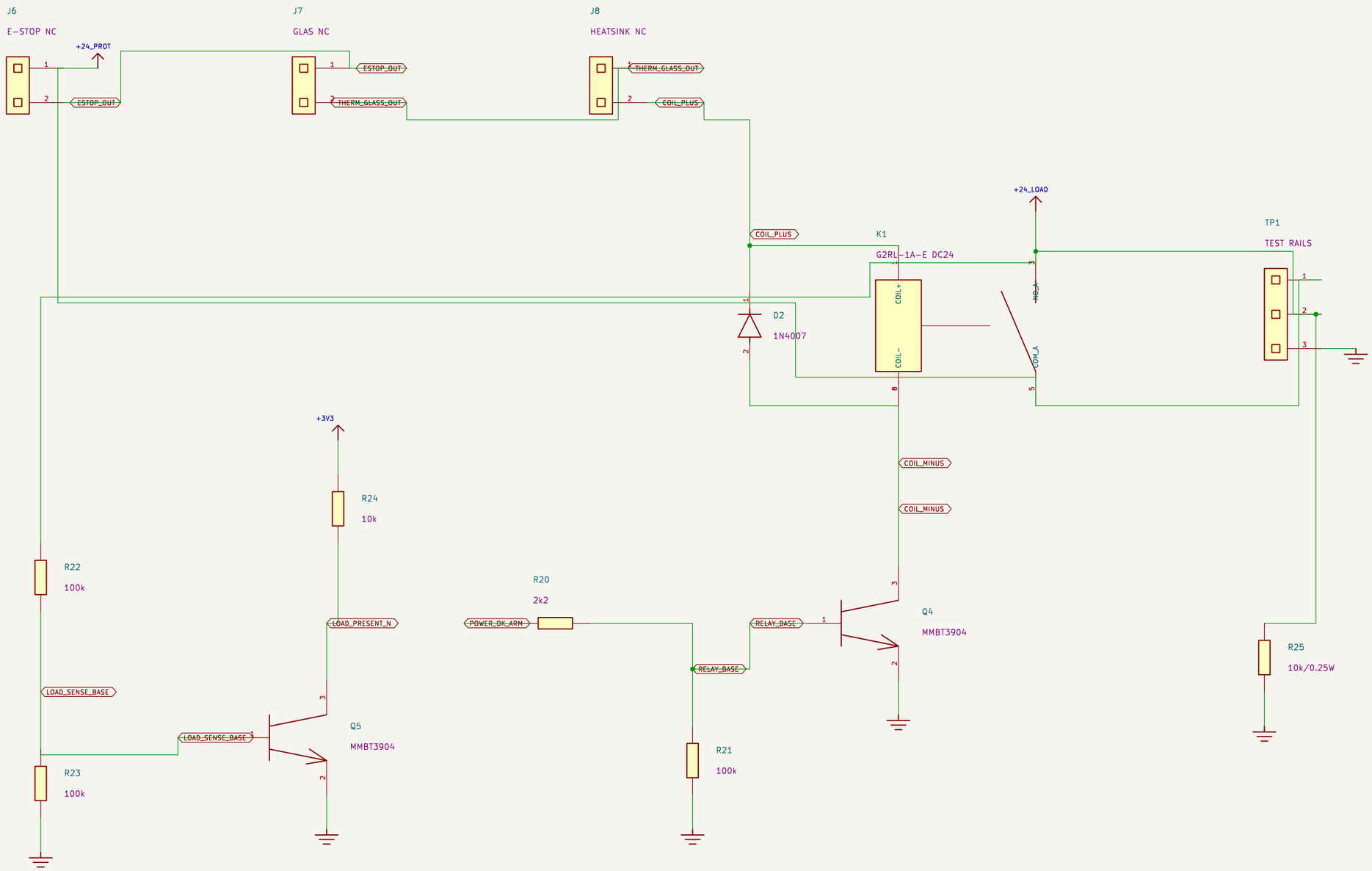
J30
R7B KPJX-4S
F1
10A
+24_FUSED
Q2
IRLB4030PBF
EFUSE_IN
U1
TPS26630RGER
J3
LAB 24V
+24_IN
D1
SMBJ33CA bidir
C6
1uF/50V X7R
PROT_GATE
Q3
BSS138LT1G
+24_FUSED
R10
156k 0.1%
R12
223k 0.1%
UV_DIV
OV_DIV
R11
10k 0.1%
R13
10k 0.1%
J9
EFUSE RESET
PROT_SHDN
IN_SYS
IN
B_GATE
DRV
UVLO
OVP
SHDN
MODE
NC
GND
OUT
PGOOD
PGTH
FLT
IMON
ILIM
dVdT
+24_PROT
PROT_PGOOD
PROT_FAULT_N
INPUT_IMON
PROT_ILIM
PROT_DVDT
R18
1k
C5
470uF/50V
+3V3
R15
10k
R16
150k 0.1%
PROT_PGTH
R17
10k 0.1%
C4
100nF/50V X7R
R14
4.02k 0.1%
U15
SN74AHC1G08DBVR
POWER_ARM
PROT_PGOOD
POWER_OK_ARM
R19
100k
C7
100nF/10V

U1: IN 1/2, OUT 17/18 en GND 8/25 zijn samengevoegd.

MODE 11 en NC 19–24 blijven open.

03 Relais en hardware-uitschakeling

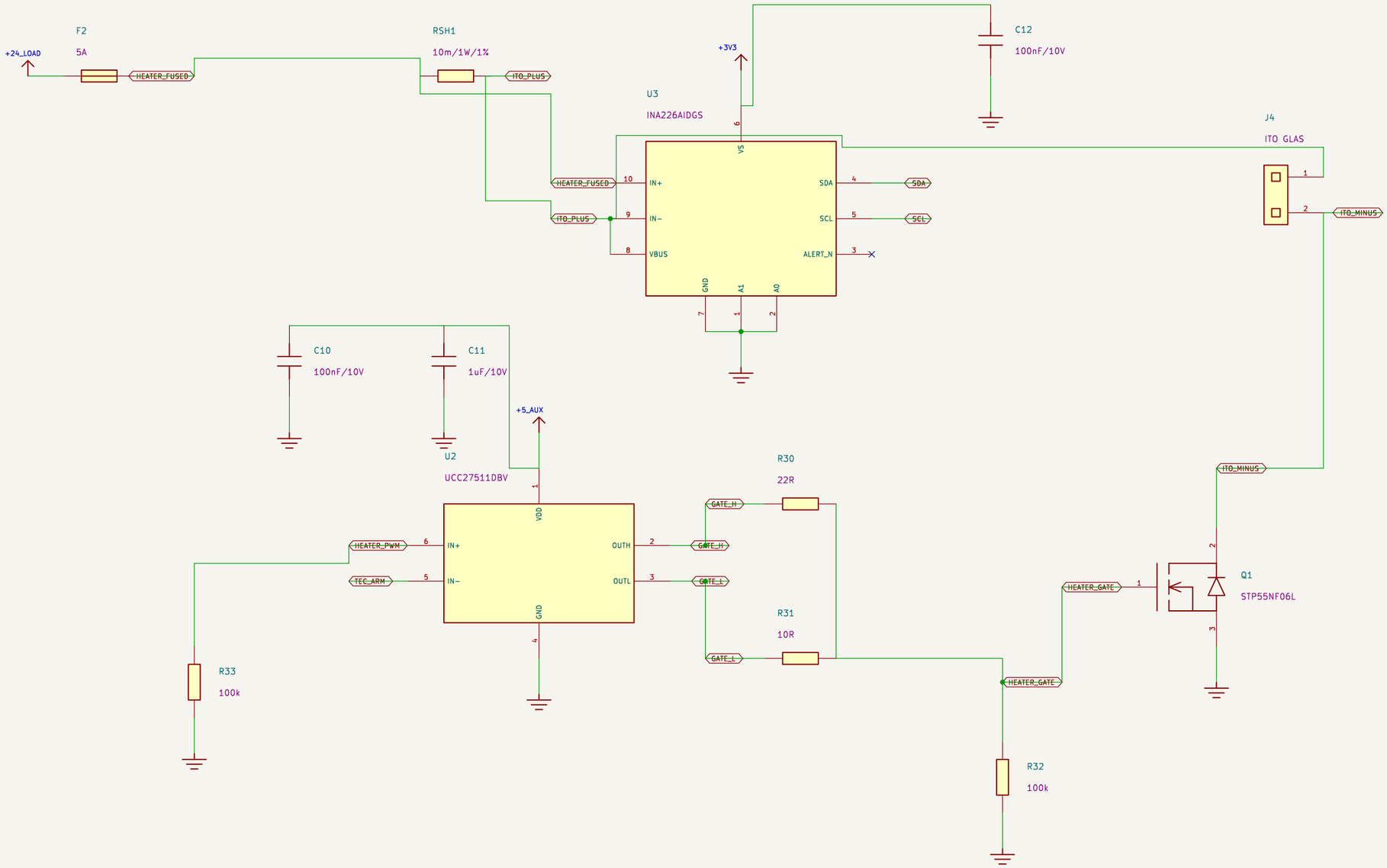
ITO controller L | 24 V | prototype 4 A | bedrading volgens bestaande PCB



K1: COM 5/6 en NO 3/4 parallel; spoel op 1/8.

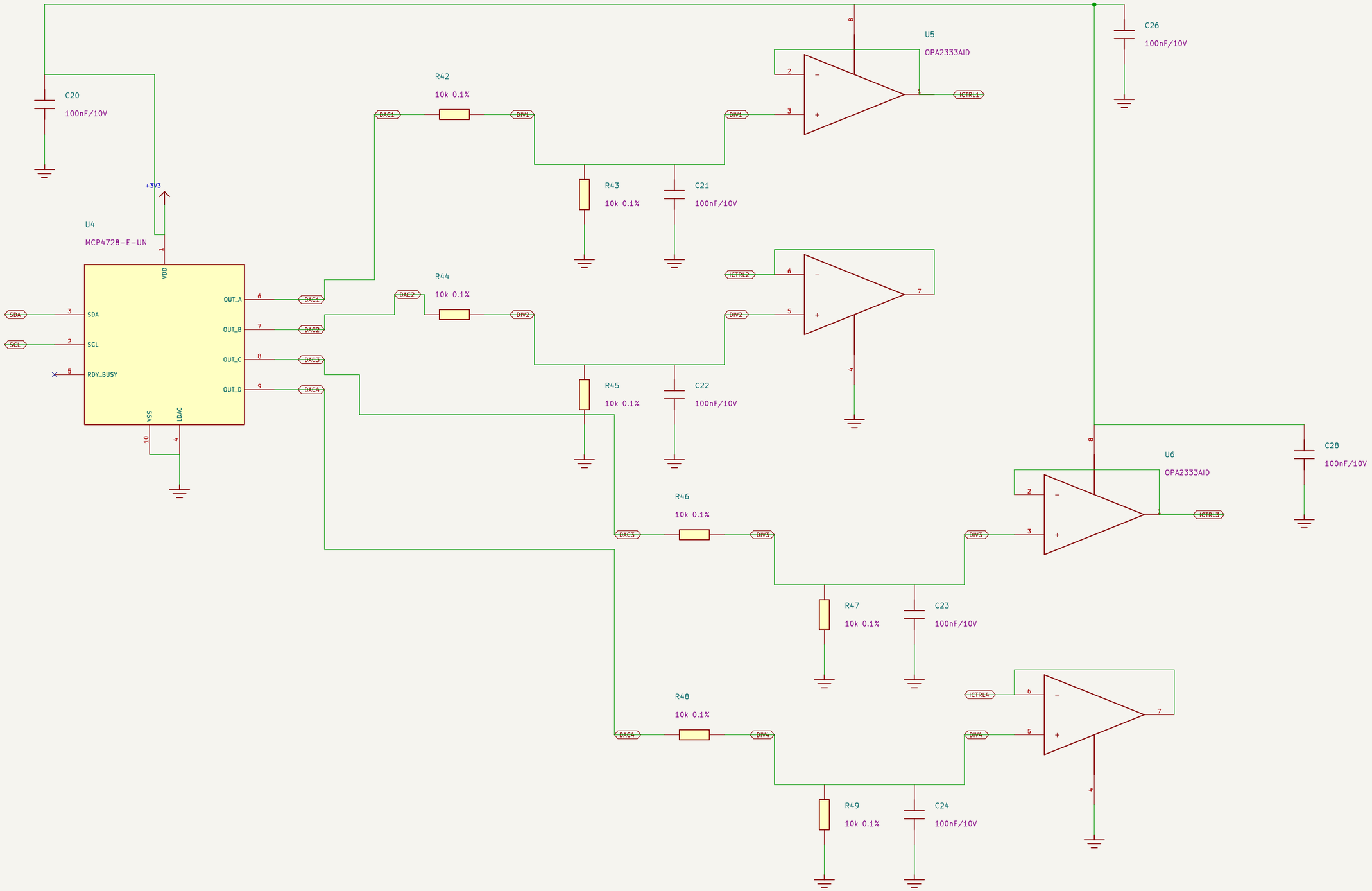
04 ITO heater

ITO controller L | 24 V | prototype 4 A | bedrading volgens bestaande PCB



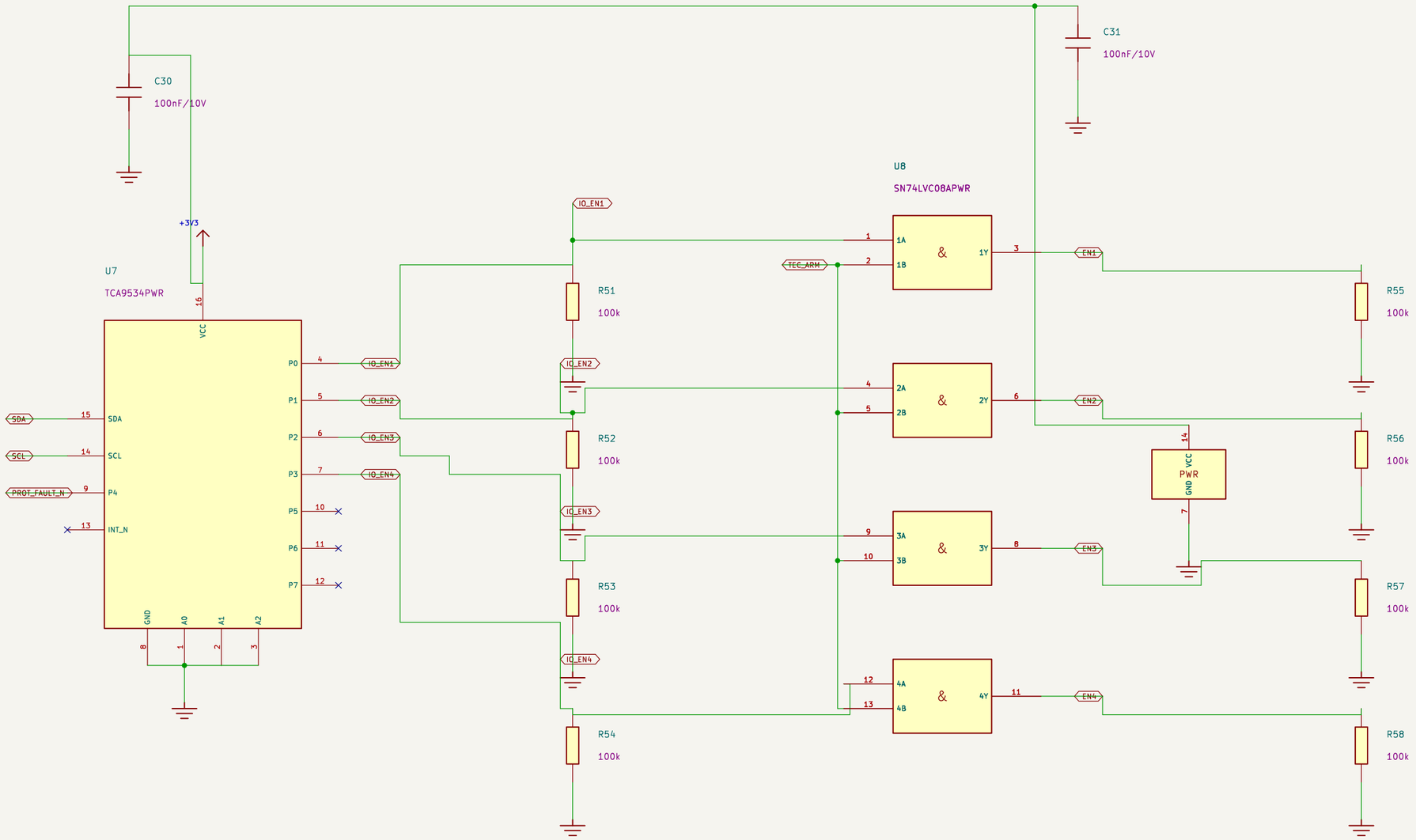
05 DAC en stroominstellingen

ITO controller L | 24 V | prototype 4 A | bedrading volgens bestaande PCB



06 Vrijgave van de Peltierkanalen

ITO controller L | 24 V | prototype 4 A | bedrading volgens bestaande PCB

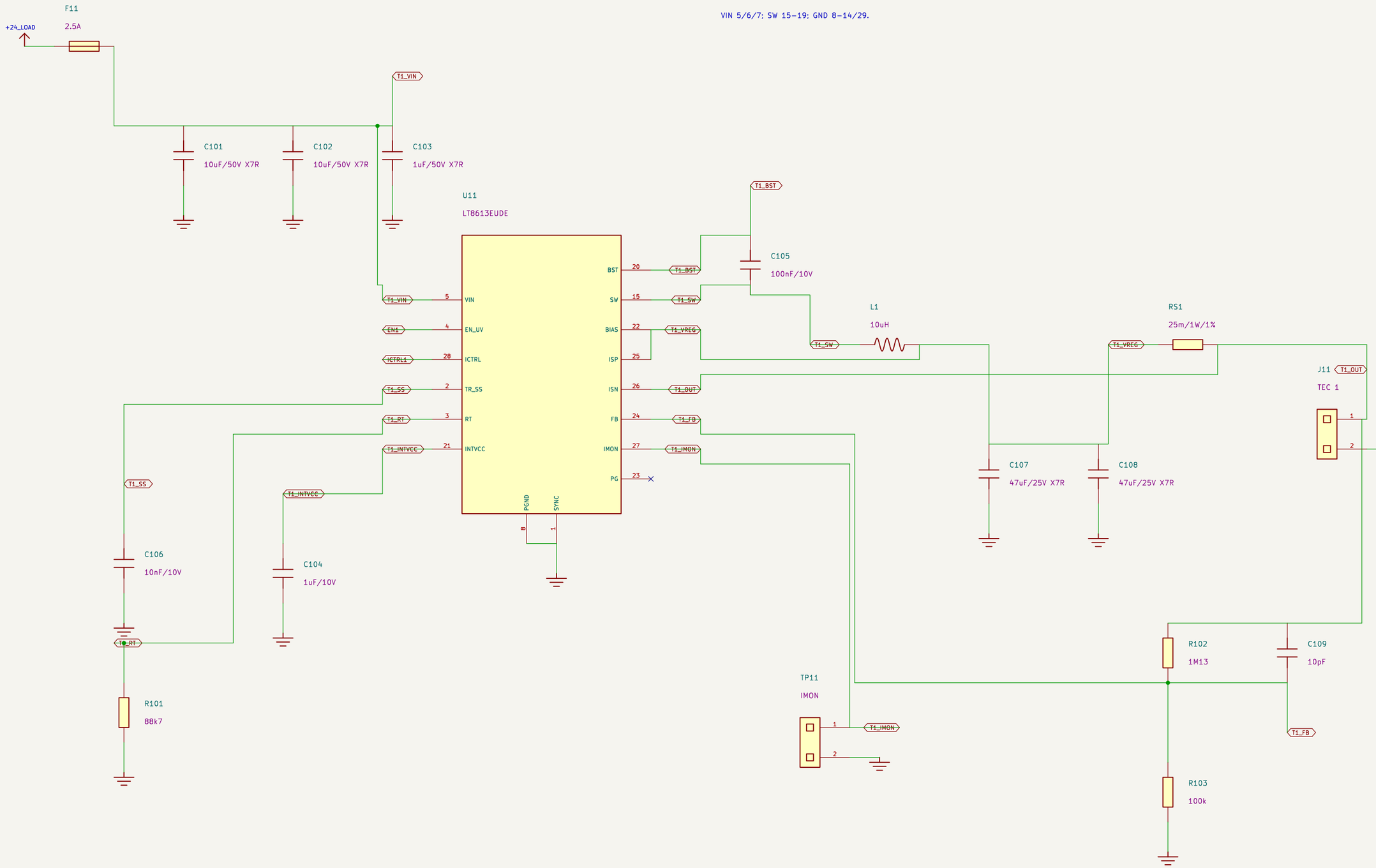


07 Peltier stroomregelaar 1

ITO controller L | 24 V | prototype 4 A | bedrading volgens bestaande PCB

Gegroepeerde IC-pinnen:

VIN 5/6/7; SW 15–19; GND 8–14/29.

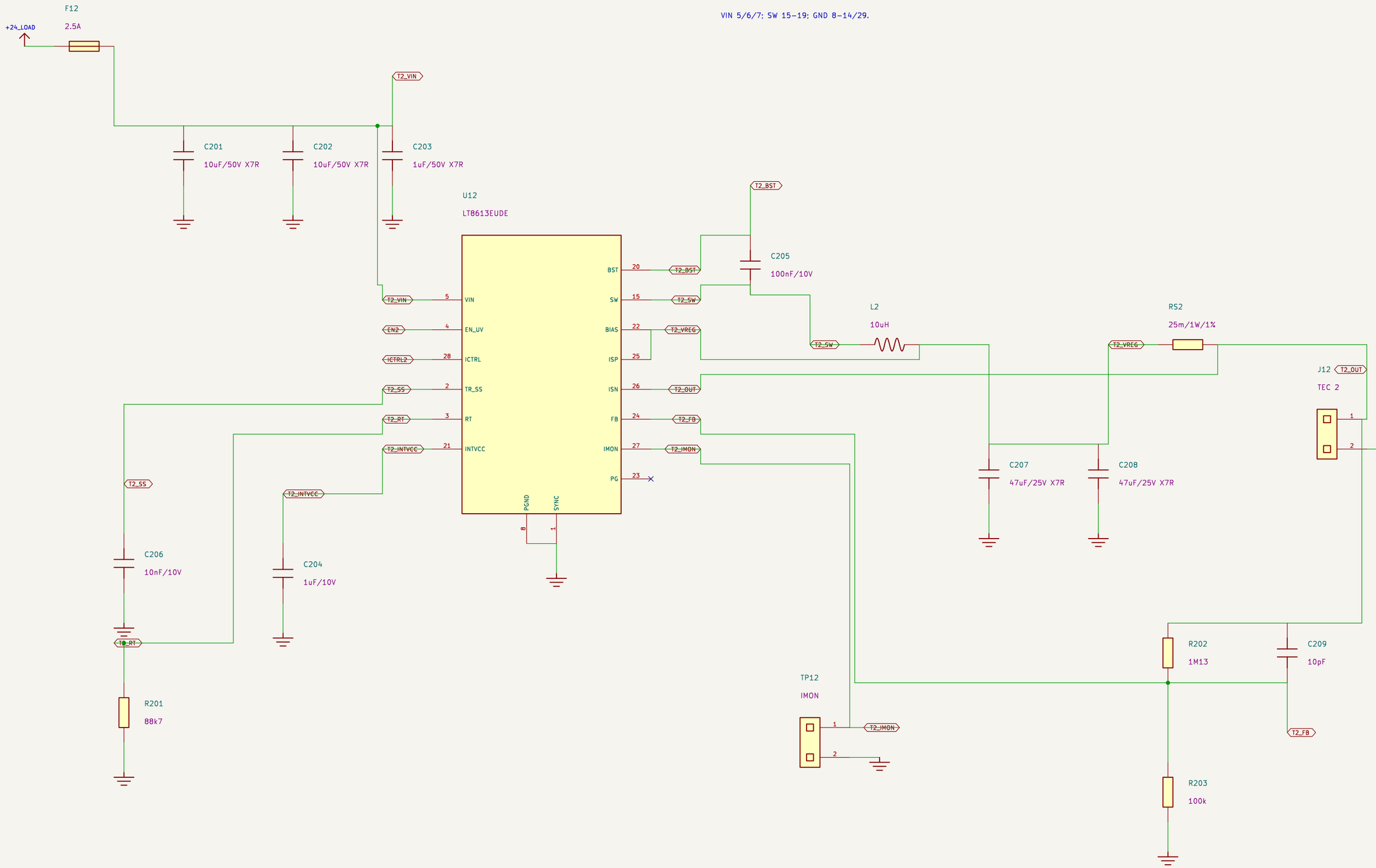


08 Peltier stroomregelaar 2

ITO controller L | 24 V | prototype 4 A | bedrading volgens bestaande PCB

Gegroepeerde IC-pinnen:

VIN 5/6/7; SW 15–19; GND 8–14/29.

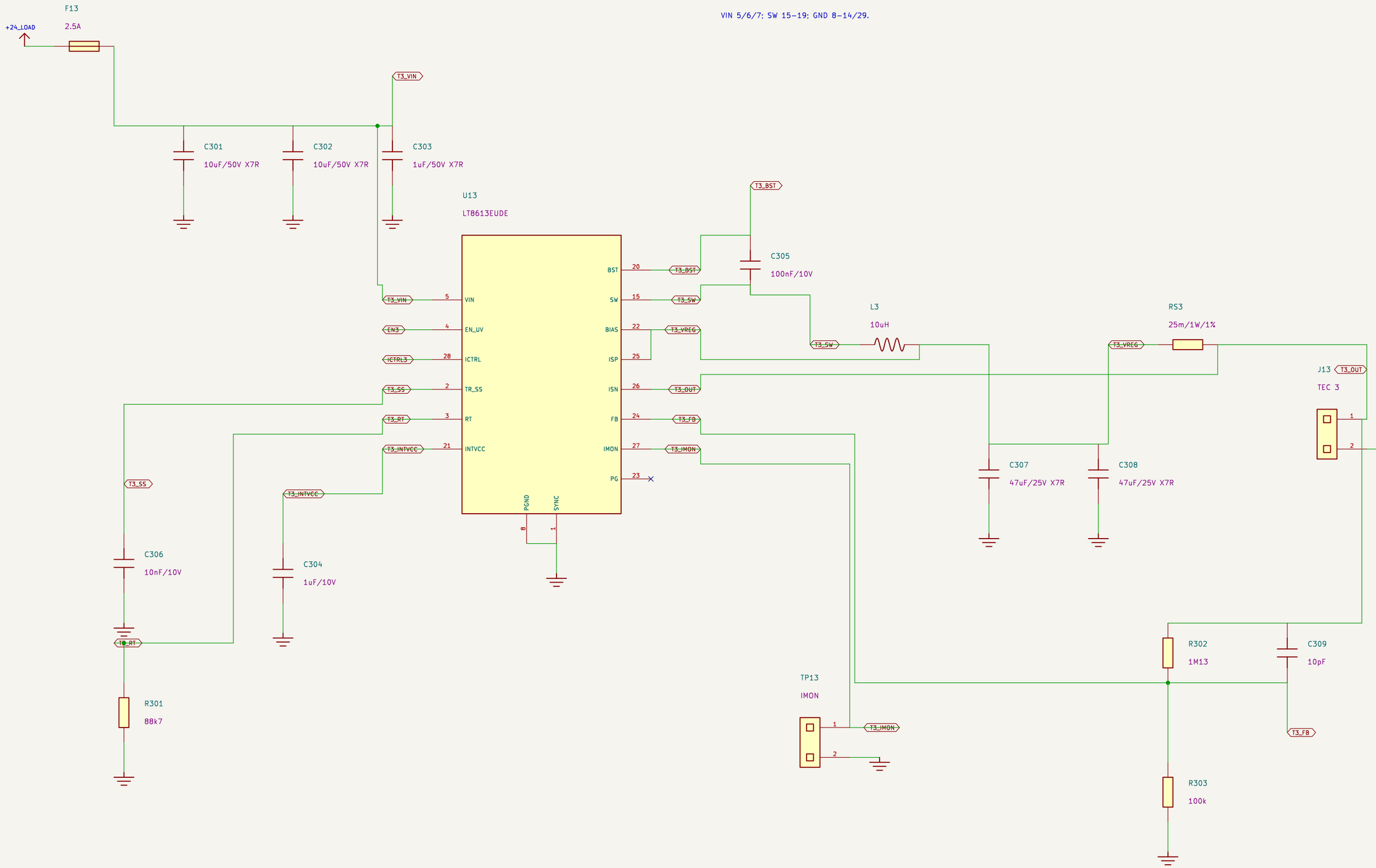


09 Peltier stroomregelaar 3

ITO controller L | 24 V | prototype 4 A | bedrading volgens bestaande PCB

Gegroepeerde IC-pinnen:

VIN 5/6/7; SW 15–19; GND 8–14/29.

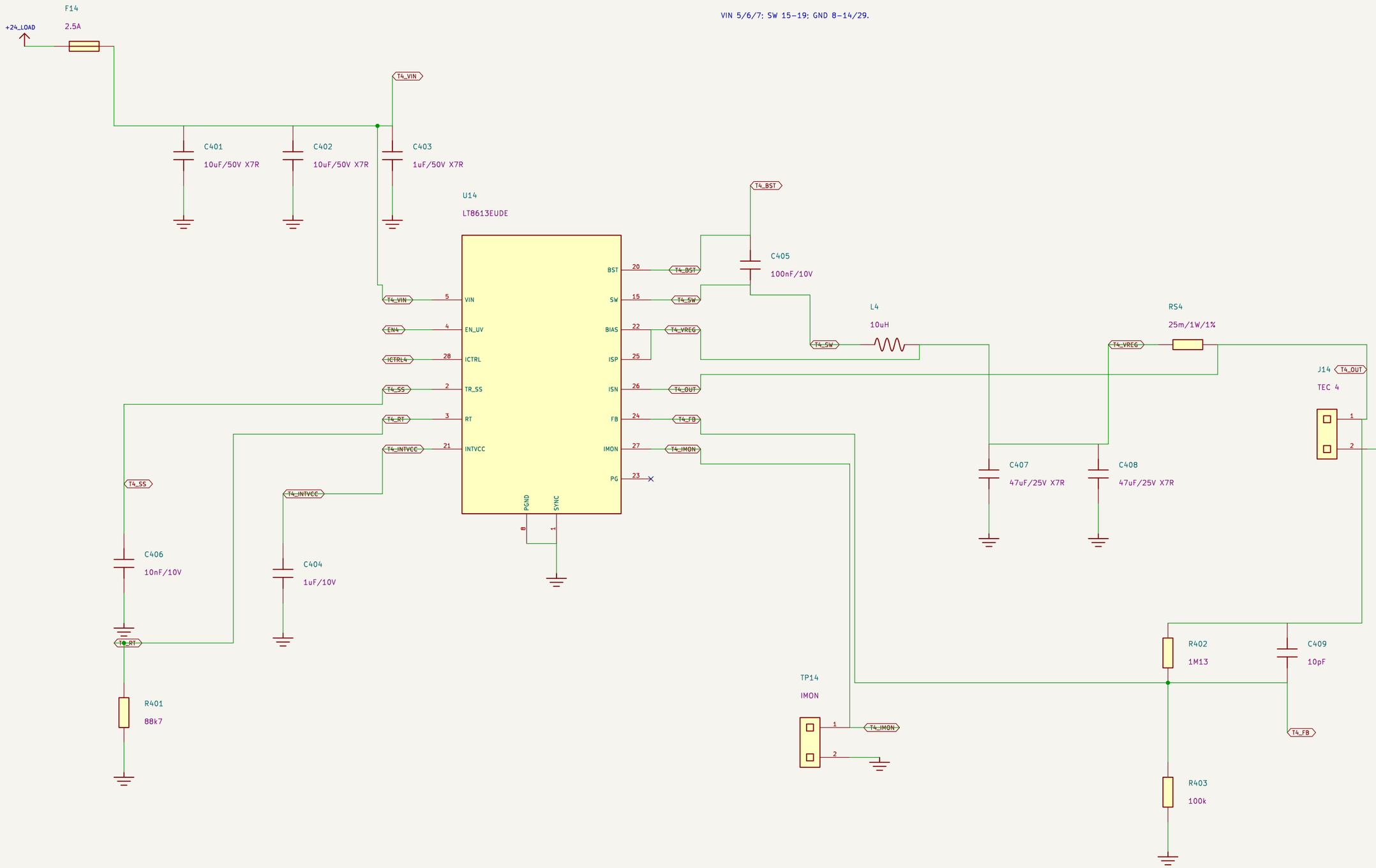


10 Peltier stroomregelaar 4

ITO controller L | 24 V | prototype 4 A | bedrading volgens bestaande PCB

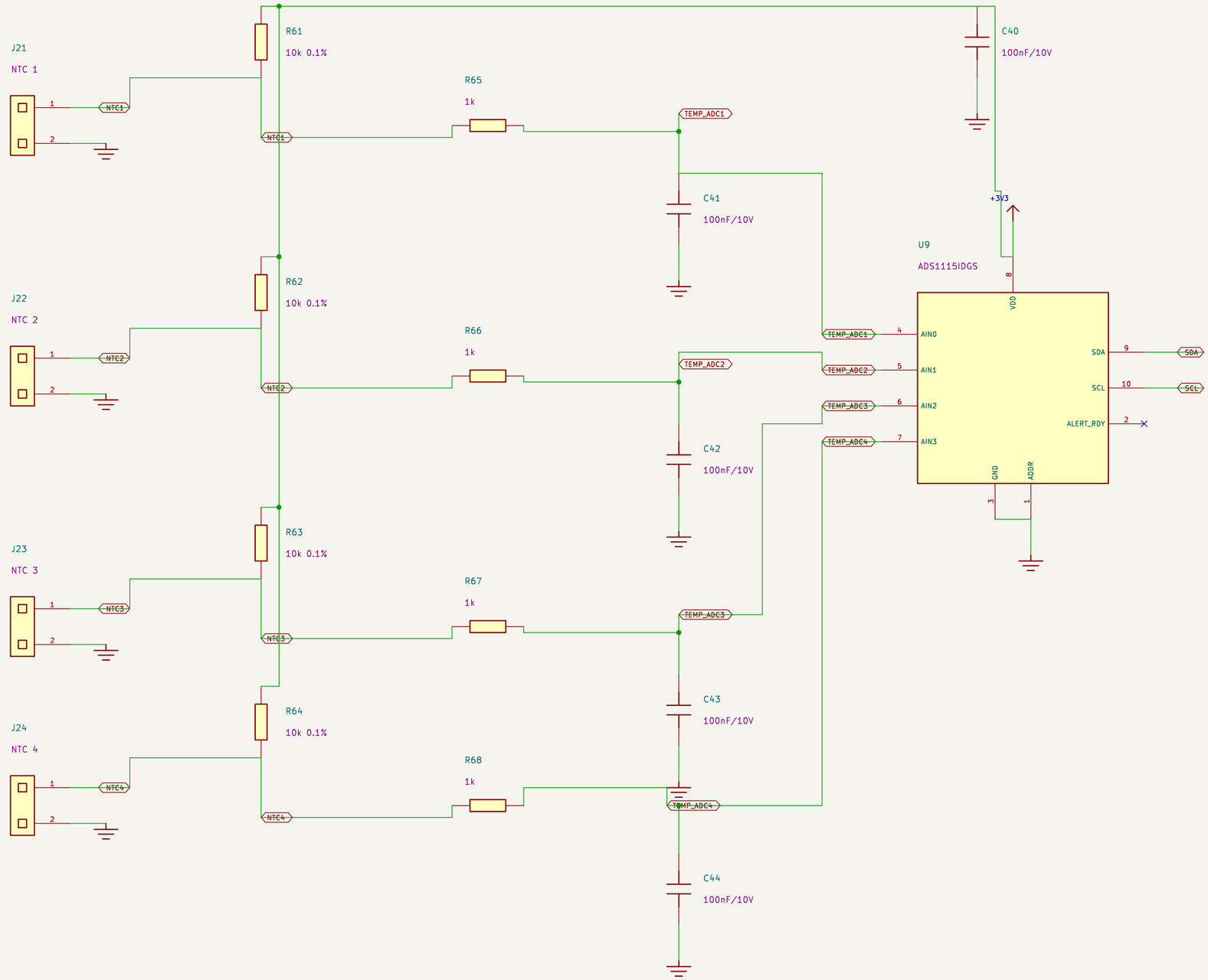
Gegroepeerde IC-pinnen:

VIN 5/6/7; SW 15–19; GND 8–14/29.



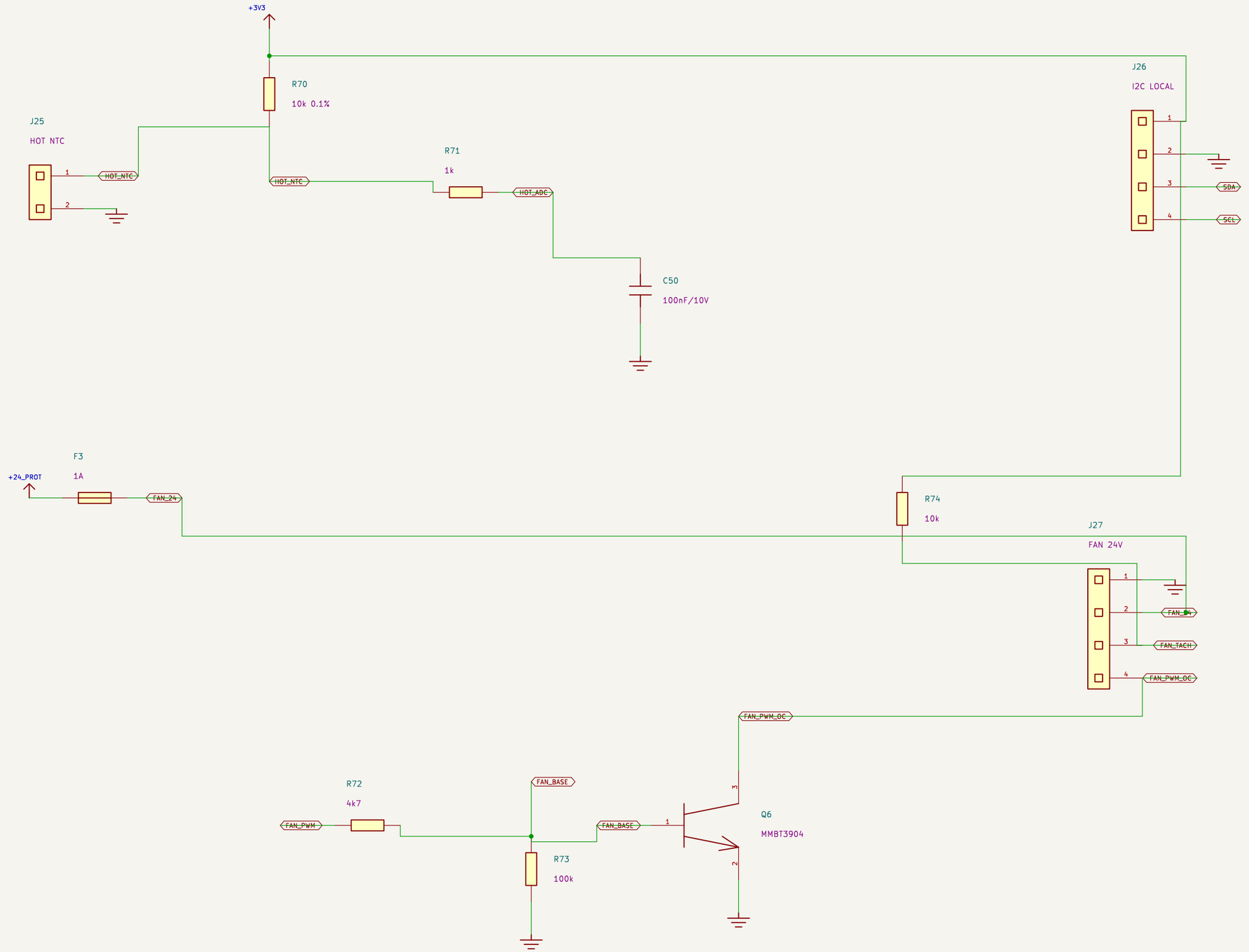
11 Temperatuurmeting op het glas

ITO controller L | 24 V | prototype 4 A | bedrading volgens bestaande PCB



12 Heatsink, luchtvochtigheid en ventilator

ITO controller L | 24 V | prototype 4 A | bedrading volgens bestaande PCB



13 Remote ambient sensor ports

ITO controller L | 24 V | prototype 4 A | bedrading volgens bestaande PCB

